

OceanEmbed Final Architecture Specification

OceanEmbed takes a temporal sequence of 7 two-dimensional surface ocean/atmospheric variables over the North Indian Ocean and reconstructs a three-dimensional subsurface ocean temperature field across 15 depth levels down to 1000 m.

The system's final input/output tensor formulation maps an input sequence (B, T, 7, 101, 241) to a reconstructed target sequence (B, T, 15, 101, 241).

1. Source of Truth & Final Architecture Decision

The final architecture is **ConvFormer**. The architecture search is closed.

This specification distinguishes between the original intended design (Architecture and Experimentation Plan) and the final established Phase-1 implementation (Research Analysis). Where differences exist, the proven Phase-1 configuration is the final specification unless explicitly marked otherwise. STGAT was evaluated as an alternative hypothesis but is rejected for final implementation.

2. Input Specification

The system consumes 7 distinct surface channels.

Channel	Variable Name	Full Name	Physical Meaning	Units	Original Source	Original Spatial Res.	Original Temporal	Dimensions Selected	Resampling Operation	Unit Conversion	Missing Value Behavior	Normalization	Final Tensor Channel	Why It Is Useful
0	sst	Sea Surface Temperature	Surface thermal energy	°C	OSTIA	0.05°	Daily	2D Surface	Spatial Interpolation	\$K \righ tarrow ^\ cir	Land mask applied	Train-only (mean/std)	0	Primary thermal core

		ra tu re	er gy						n	c C \$		d)		ns tr ai nt fo r su bs ur fa ce pr ofi le
1	ss s	Se a Su rf ac e Sa lini ty	Fr es h w at er bo un da ry	ps u	S M A P/ S M O S	0. 25 °	D ail y	D ep th =1 sq ue ez ed to 2D	N on e	N on e	La nd m as k ap pli ed	Tr ai n- on ly (m ea n/ st d)	1	Re so lv es fr es h w at er ba rri er la ye rs (e. g., Ba y of Be ng al)
2	sla	Se	Th	m	C	0.	D	2D	C	N	La	Tr	2	In

		a Le ve l A no m al y	er m oc lin e pr ox y		M E M S S L A	25 °	ail y	Su rf ac e	ell -c en te r of fs et re gri dd ed	on e	nd m as k ap pli ed	ai n- on ly (m ea n/ st d)		di ca te s su bs ur fa ce m as s/ he at an o m ali es (e dd ie s, up w elli ng)
3	u	Zo na l Su rf ac e C ur re nt	E/ W su rf ac e tr an sp or t	m/ s	C M E M S C ur re nt s	0. 25 °	D ail y	D ep th =0 se le ct ed	C ell -c en te r of fs et re gri dd	N on e	La nd m as k ap pli ed	Tr ai n- on ly (m ea n/ st d)	3	C ap tu re s ad ve cti on an d eq ua

									ed					to ria l jet str uc tu re s
4	v	M er i di on al Su rf ac e C ur re nt	N/ S su rf ac e tr an sp or t	m/ s	C M E M S C ur re nt s	0. 25 °	D ail y	D ep th =0 se le ct ed	C ell -c en te r of fs et re gri dd ed	N on e	La nd m as k ap pli ed	Tr ai n- on ly (m ea n/ st d)	4	C ap tu re s ad ve cti on an d cr os s- eq ua to ria l flo w s
5	u1 0	ER A 5 Zo na l Wi nd	E/ W at m os ph eri c	m/ s	ER A 5	0. 25 °	D ail y	2D Su rf ac e	N on e (n ati ve ly m	N on e	G ap -fr ee (0 % mi ssi	Tr ai n- on ly (m ea n/ st d)	5	Dr iv es su rf ac e mi

			fo rci ng						at ch es)		ng)	st d)		xi ng an d Ek m an tr an sp or t
6	v1 0	ER A 5 M eri di on al Wi nd	N/ S at m os ph eri c fo rci ng	m/ s	ER A 5	0. 25 °	D ail y	2D Su rf ac e	N on e (n ati ve ly m at ch es)	N on e	G ap -fr ee (0 % mi ssi ng)	Tr ai n- on ly (m ea n/ st d)	6	Dr iv es su rf ac e mi xi ng an d Ek m an tr an sp or t

3. Final Spatial Grid

- **Latitude Region:** 5°N–30°N
 - **Longitude Region:** 45°E–105°E
 - **Resolution:** 0.25°
 - **Final Implemented Grid:** 101 × 241
- Coordinate Construction:** The original architecture plan proposed a 100 × 240 grid. However,

strict geographic mathematics at 0.25° resolution requires inclusive boundary nodes.

- **Latitude:** $(30 - 5) / 0.25 = 100$ intervals. Including the starting node at 5°N yields exactly 101 latitude points.
- **Longitude:** $(105 - 45) / 0.25 = 240$ intervals. Including the starting node at 45°E yields exactly 241 longitude points.

The system strictly utilizes the 101 × 241 tensor layout to maintain exact alignment with the bounding box.

- **Tensor Convention:** (..., H, W) corresponds to (..., Latitude, Longitude).

4. Temporal Input

- **Sequence Length:** 5 (The architecture plan proposed 7, but Phase 1 successfully implemented and validated 5. Sequence length is 5 unless actively re-tuned during the final Kaggle run).
- **Input Tensor:** (B, 5, 7, 101, 241)
 - **B:** Batch size.
 - **5:** Temporal sequence of 5 consecutive days.
 - **7:** Input variables.
 - **101:** Latitude nodes.
 - **241:** Longitude nodes.

The 5 timesteps are consecutive chronological days. The system aligns inputs such that the temporal dimension retains sequence integrity without boundary-crossing between training, validation, and test splits.

5. Target Specification

- **Target Depth Levels (15):** 0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700, 1000 m.
- **Target Source:** GLORYS12V1 Reanalysis (during training).
- **Units:** °C.
- **Target Tensor:** (B, 5, 15, 101, 241). The target matches the sequence length T=5, providing a corresponding 3D subsurface field for every input timestep.

0 m Extrapolation: GLORYS does not natively output a 0 m depth (shallowest is ~0.494 m, second is ~1.541 m). The exact 0 m surface target is generated via 1D linear extrapolation:

$$T(0) = T(0.494) + (0 - 0.494) \times \frac{T(1.541) - T(0.494)}{1.541 - 0.494}$$

This mathematically anchors the vertical profile to a physically plausible surface boundary.

6. Data Preprocessing → Model Input

The pipeline enforces strict transformations before data reaches PyTorch.

Stage	Input	Operation	Output	Shape	Missing Value Behavior

RAW INGESTION	Raw NetCDFs	Read heterogeneous sources	Extracted arrays	Variable	Preserved
DIMENSION CLEANUP	SSS array	Squeeze depth=1 axis	2D SSS array	(T, 101, 241)	Preserved
UNIT CONVERSION	SST (Kelvin)	Apply $C = K - 273.15$	SST (Celsius)	(T, 101, 241)	Preserved
HORIZONTAL REGRID	SLA, U, V	Spatial interpolation to 0.25°	Grid-aligned arrays	(T, 101, 241)	Preserved
VERTICAL INTERP	GLORYS 3D	1D interp to 15 depths + 0m extra	Target sequence	(T, 15, 101, 241)	Target NaNs preserved
OCEAN MASK	All Variables	Apply bounding mask	Masked arrays	(T, C, 101, 241)	Land set to NaN
CHANNEL STACK	7 variables	Stack along channel dimension	Combined Input	(T, 7, 101, 241)	NaNs preserved
NORMALIZATION	Combined Input	Apply Train-Only Mean/Std	Normalized Input	(T, 7, 101, 241)	NaNs persist temporarily
INPUT NaN FILL	Normalized Input	Convert Input NaNs to 0.0	Clean Input	(T, 7, 101, 241)	Input NaNs filled with 0.0
SEQUENCE EXTRACT	Clean Input / Target	Windowing (stride=1)	PyTorch Tensors	(B, 5, 7, 101, 241)	0.0 (Input), NaN (Target)

7. ERA5 vs ASCAT

The final atmospheric forcing relies entirely on **ERA5 U10 and V10** reanalysis. The initially proposed ASCAT satellite winds were rejected due to extreme missingness (only 16.88% valid data). Resolving ASCAT required massive nearest-neighbor spatial imputation resulting in a mean imputation distance of 237.37 km and a P95 distance of 500.27 km. This artificial spatial smearing destroyed high-frequency atmospheric variance. ERA5 guarantees a gap-free forcing field without spatial smearing and maps perfectly to the 0.25° grid.

8. Normalization

Normalization is strictly constrained to prevent data leakage.

- **Formula:** $x_{\text{norm}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$
- **Statistics:** Computed globally per-channel, **exclusively on the training split**.
- **Validation/Test:** Must apply the frozen training statistics. They must never be recomputed on validation/test data.
- **Storage:** normalization_stats_train.json.

9. Masking and NaN Handling

Input and Target tensors handle missingness differently to satisfy both PyTorch layer requirements and physical truth.

- **Input NaNs:** Deep learning convolutions propagate NaNs, destroying the gradient. Therefore, all geographic land/invalid input pixels are explicitly filled with 0.0 **after** normalization.
- **Target NaNs (Seafloor/Land Mask):** Target missingness represents locations where the physical ocean depth is undefined. These must **not** be globally imputed because training the model to predict pseudo-temperatures inside solid earth induces meaningless gradients.
- **Valid Target Mask:** Defined mathematically as `~torch.isnan(target)`.
- **Application:** The loss function is computed exclusively where this mask evaluates to True.

10. Complete ConvFormer Architecture

The network is a Convolutional Vision Transformer hybrid.

Stage	Operation	Input Shape	Output Shape	Parameters	Purpose
1. Feature Extractor	Spatial CNN	(B, 5, 7, 101, 241)	(B, 5, 64, 101, 241)	TO BE CONFIRMED	Extracts local spatial gradients (fronts, eddies)

2. Temporal Block	ConvLSTM	(B, 5, 64, 101, 241)	(B, 5, 64, 101, 241)	TO BE CONFIRMED	Models chronological evolution over 5 days
3. Patch Embedding	Conv2d (stride)	(B, 5, 64, 101, 241)	(B, 5, C_emb, H', W')	TO BE CONFIRMED	Compresses spatial grid into tokenizable patches
4. Flatten & PE	Reshape + Add PE	(B, 5, C_emb, H', W')	(B, 5, N, D)	TO BE CONFIRMED	Prepares sequence for self-attention
5. Transformer	Encoder Blocks	(B, 5, N, D)	(B, 5, N, D)	TO BE CONFIRMED	Captures global basin teleconnections
6. Reshape	Unflatten	(B, 5, N, D)	(B, 5, C_emb, H', W')	0	Restores spatial layout
7. Decoder	Shared Decoder	(B, 5, C_emb, H', W')	(B, 5, C_dec, 101, 241)	TO BE CONFIRMED	Upsamples back to original 0.25° grid
8. Projection	1x1 Conv	(B, 5, C_dec, 101, 241)	(B, 5, 15, 101, 241)	TO BE CONFIRMED	Maps features to the 15 specific physical depths

11. CNN Spatial Feature Extractor

- **Input:** (B, 5, 7, 101, 241)
- **Output:** (B, 5, C_emb, 101, 241) (Proposed C_emb = 64).
- **Operation:** Time-distributed 2D convolutions (applied per timestep) mapping raw physical variables into latent spatial feature maps.
- **Detail:** *Exact layer count, kernel sizes, and padding: TO BE CONFIRMED FROM IMPLEMENTATION.* The grid resolution (101x241) must be preserved at this stage.

12. ConvLSTM

- **Input:** (B, 5, C_emb, 101, 241)
- **Output:** (B, 5, C_emb, 101, 241) (All timesteps are retained for sequence-to-sequence prediction).
- **Proposed Kernel:** 3x3.
- **Purpose:** The recurrent ConvLSTM block embeds temporal flow natively, capturing historical persistence while retaining 2D spatial locality.

13. Patch Embedding

- **Input:** (B, 5, C_emb, 101, 241)
- **Operation:** 4x4 Convolutional Kernel with Stride 4.
- **Grid Mathematics:** A 101×241 grid is not cleanly divisible by 4. Therefore, exact padding (e.g., to 104×244) or cropping logic is strictly required prior to patch extraction. *Exact padding method: TO BE CONFIRMED FROM IMPLEMENTATION.*
- **Tokens (N):** Assuming standard padding to 104×244 , $N = (104/4) \times (244/4) = 26 \times 61 = 1586$ spatial tokens.
- **Output:** (B, 5, N, D) where D is the embedding dimension.

14. Token Representation & Positional Encoding

- **Flattening:** The spatial tensor is flattened along the height and width dimensions to yield a sequence of length N.
- **Positional Encoding:** Standard vision transformers lack spatial awareness. A positional encoding (PE) tensor is added to the flattened embeddings. *PE formulation (Sinusoidal vs. Learned): TO BE CONFIRMED FROM IMPLEMENTATION.*

15. Transformer Encoder

- **Input Sequence:** (B, 5, N, D)
- **Blocks:** Proposed L=4.
- **Operation:** Multi-head self-attention applied globally across spatial patches.
- **Mechanism:** Queries, Keys, and Values are derived from the tokenized patches. Attention is calculated dynamically, allowing Patch i (e.g., Equatorial region) to directly influence Patch j (e.g., Bay of Bengal) regardless of grid distance. *Exact number of heads and feed-forward dimensions: TO BE CONFIRMED FROM IMPLEMENTATION.*

16. Temporal vs Spatial Modeling (Physics Mapping)

- **CNN:** Captures local advection, thermal fronts, and mesoscale eddies.

- **ConvLSTM:** Tracks recent temporal persistence and dynamic heating/cooling trends over the 5-day window.
- **Patch Embedding:** Provides spatial compression, grouping local physics into regional nodes.
- **Transformer:** Models global basin teleconnections, allowing distant physical drivers (e.g., equatorial winds) to inform remote subsurface responses.
- **Depth Projection:** Reconstructs the vertical gradient, specifically resolving the thermocline.

17. Decoder

- **Input:** (B, 5, N, D)
- **Reshape:** Unflattened back to (B, 5, C_emb, H', W').
- **Upsampling:** A spatial upsampling module (e.g., ConvTranspose2d) scales the patch grid back to the original 101×241 resolution. *Exact upsampling layers/skip connections: TO BE CONFIRMED FROM IMPLEMENTATION.*

18. Depth Projection

- **Input:** Upsampled latent features at 101×241 .
- **Operation:** A terminal 1×1 convolutional layer.
- **Output:** Maps the latent channels directly to the 15 distinct depth target channels, outputting the final (B, 5, 15, 101, 241) tensor.

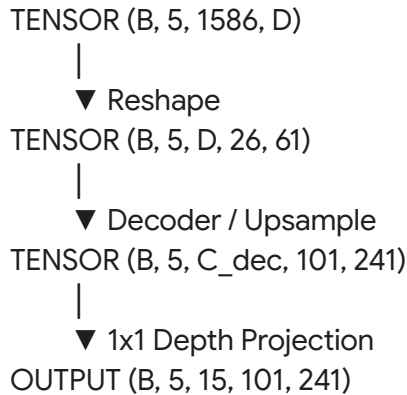
19. Complete Forward Pass Tensor Flow

Code snippet

```

INPUT (B, 5, 7, 101, 241)
|
▼ CNN Feature Extractor
TENSOR (B, 5, 64, 101, 241)
|
▼ ConvLSTM
TENSOR (B, 5, 64, 101, 241)
|
▼ Patch Embed (4x4, padded)
TENSOR (B, 5, D, 26, 61) <-- H'/W' dimensions based on padding
|
▼ Flatten & PE
TENSOR (B, 5, 1586, D)
|
▼ Transformer Encoder (L=4)

```



20. Parameter Count

- **Total ConvFormer Parameters:** 1,064,847.
- **Component Breakdown:** NOT CONFIRMED FROM AVAILABLE IMPLEMENTATION.

21. Loss Function

- **Actual Implemented Training Loss:** Masked MSE.
- **Formulation:**

$$L_{\text{MSE}} = \frac{1}{\sum_i M_i} \sum_i \left(M_i \cdot (Y_i - \hat{Y}_i)^2 \right)$$
 Where M is the valid target mask ($\sim \text{torch.isnan}(\text{target})$). Invalid cells explicitly yield a gradient of 0.
- **Evaluation Metrics:** Masked MAE, Masked RMSE, and R^2 .
- **PROPOSED / FUTURE:** Density Stability Penalty ($\lambda \mathcal{L}_{\text{density}}$) to penalize thermodynamic inversions. (Not confirmed in Phase-1 implementation).

22. Training Configuration

Parameter	Phase-1 Implementation PDF	Final Multi-Year Setting
Framework	PyTorch	PyTorch
Optimizer	AdamW	AdamW
Learning Rate	1e-3	1e-3 (Configurable)
Weight Decay	1e-4	1e-4 (Configurable)
Scheduler	ReduceLROnPlateau	ReduceLROnPlateau

Early Stopping	8 epochs	Configurable
Max Epochs	50	Configurable for Kaggle
Seed	42	42
Batch Size	1 (Local Mac)	TO BE CONFIRMED (Kaggle limit)

23. Checkpoint Content

The training loop saves an exhaustive .pt checkpoint to guarantee reproducibility:

- `model_state_dict` (architecture weights)
- `optimizer_state_dict` (momentum, step counts)
- `scheduler_state_dict` (learning rate state)
- `epoch` (current progress)
- `best_val_loss` (metric to beat)
- `model_configuration` (ensures exact shape reproducibility)
- `seed` (random state)

24. Final Multi-Year Data Flow

The final chronologically strict dataset partitioning prevents data leakage.

- **Training Set (1993–2018):** Target is GLORYS reanalysis.
- **Validation Set (2019–2020):** Target is GLORYS reanalysis (used exclusively for early stopping).
- **Independent Evaluation (2021–2023):** Target is INCOIS Gridded ARGO (in-situ independent ground truth, strictly held out until the final evaluation).

25. Large-Data Training Architecture

Because 30 years of daily arrays will exceed standard RAM limits, the Kaggle architecture requires:

1. **Chunked Preprocessing:** Raw data is converted into chunked Zarr arrays or chunked NetCDF files.
2. **PyTorch Dataset:** Implements lazy-loading using xarray or zarr, loading only the requested T=5 index slices into memory.
3. **DataLoader:** Utilizes multiprocessing workers to prefetch batches.

26. Inference Architecture

Inference executes a precise forward pass on new data:

1. **Input:** New un-normalized sequence of 7 variables.
2. **Validation:** Ensure bounds match 101x241, 5°N–30°N, 45°E–105°E.
3. **Normalization:** Apply the frozen `normalization_stats_train.json` variables.
4. **NaN Fill:** Set land pixels to 0.0.

5. **Forward Pass:** Pass (1, 5, 7, 101, 241) tensor through ConvFormer.
6. **Output:** Yields (1, 5, 15, 101, 241) tensor.
7. **Derived Metrics:** Subsurface slices are passed to the downstream D26 calculator.

27. D26 / TCHP Derivation

The model outputs a 15-depth temperature vector for every geographic coordinate. To compute Tropical Cyclone Heat Potential (TCHP), the depth of the 26°C isotherm (D_{26}) must be determined.

1. Scan the predicted vertical profile from 0 m downwards.
2. Find the consecutive depth levels z_1 and z_2 where the temperature drops below 26°C (i.e., $T_1 \geq 26^\circ\text{C}$ and $T_2 < 26^\circ\text{C}$).
3. Linearly interpolate the exact depth:

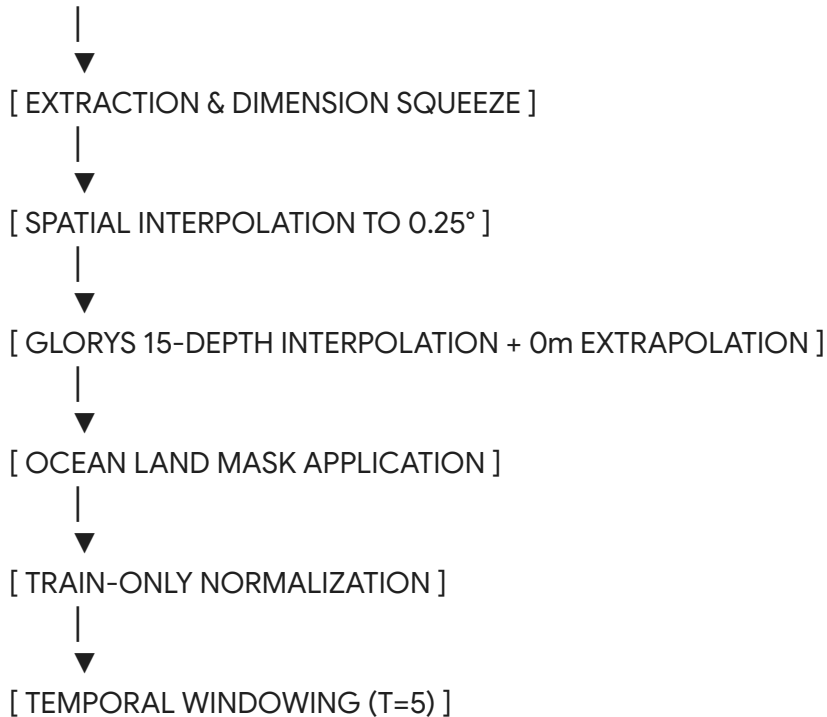
$$D_{26} = z_1 + (26 - T_1) \times \frac{z_2 - z_1}{T_2 - T_1}$$
 Edge Cases: If the surface is $< 26^\circ\text{C}$, D_{26} is NaN. If the temperature remains $> 26^\circ\text{C}$ to 1000m, $D_{26} > 1000$ m.

28. End-to-End Architecture Diagrams

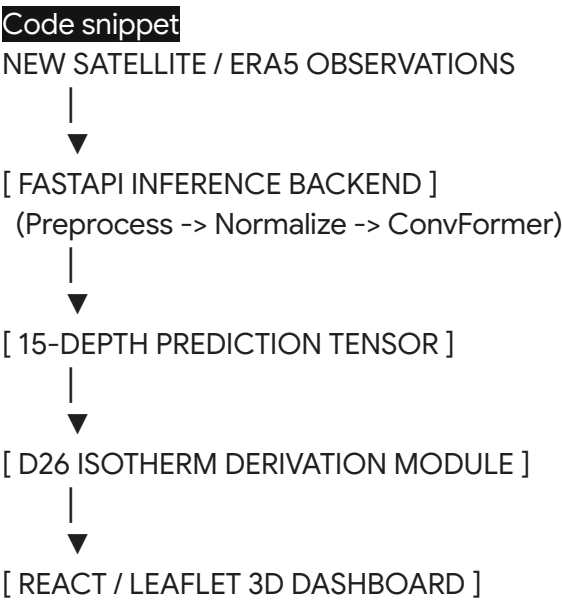
A. DATA PIPELINE

Code snippet

RAW NETCDF (SST, SSS, SLA, U, V, ERA5_U, ERA5_V, GLORYS)



B. DEPLOYMENT PIPELINE



29. Fixed vs Configurable Requirements

FIXED (Strictly enforced by Architecture)	CONFIGURABLE (Can be tuned in Kaggle)
Geographic Domain (5°N–30°N, 45°E–105°E)	Training Batch Size
Spatial Tensor Grid (101 × 241)	Number of Epochs
Input Variables (The 7 Channels)	Learning Rate (Default 1e-3)
Target Depths (The 15 Levels)	Early Stopping Patience
Normalization Method (Train-only stats)	DataLoader Chunk Size / Workers
Loss Function (Valid-Masked MSE)	Mixed Precision (FP16/FP32)
Chronological Data Splits	Checkpoint Export Frequency

30. Implementation Contract

MODEL INPUT:

- Shape: (B, 5, 7, 101, 241).
- Type: PyTorch Float32 Tensor.
- State: Standard-scaled (mean=0, std=1) based on 1993-2018 training stats.
- Missing Data: Land input pixels strictly equal 0.0.

MODEL OUTPUT:

- Shape: (B, 5, 15, 101, 241).
- Type: PyTorch Float32 Tensor.
- State: Un-normalized (native Celsius output).
- Depth Ordering: Strictly 0, 5, 10, 20, 30, 50, 75, 100, 125, 150, 200, 300, 500, 700, 1000 m.

PREPROCESSOR CONTRACT: The data engineering pipeline guarantees that data entering the model aligns precisely with the 101x241 grid and contains NO input NaNs.

MODEL CONTRACT: The ConvFormer guarantees it will execute a spatial-temporal sequence forward pass and return 15 channels corresponding exactly to the target physical depths.

POSTPROCESSOR CONTRACT: The backend guarantees it will handle D26 derivation downstream of the neural network, treating the model output strictly as a temperature volume.

31. Final Architecture Summary

- **Project:** OceanEmbed
- **Region:** North Indian Ocean (5°N–30°N, 45°E–105°E)
- **Grid:** 101 × 241 at 0.25° resolution
- **Input:** 7 surface variables (SST, SSS, SLA, U, V, ERA5 U, ERA5 V)
- **Temporal Sequence:** 5 consecutive timesteps
- **Model:** ConvFormer (Convolutional Vision Transformer hybrid)
- **Output:** 15 subsurface temperature depths
- **Depth Range:** 0 m to 1000 m
- **Loss:** Valid-Masked MSE
- **Primary Metrics:** RMSE, MAE, R²
- **Final Selected Model:** ConvFormer